

Лабораторная работа

Номер 3

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01 января 1970

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Информация

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Изучение посредством Wireshark кадров Ethernet, анализ PDU протоколов транспортного и прикладного уровней стека TCP/IP.

```
C:\Users\denis>ipconfig

Настройка протокола IP для Windows

Неизвестный адаптер Подключение по локальной сети:

    Состояние среды. . . . . : Среда передачи недоступна.
    DNS-суффикс подключения . . . . . :

Адаптер Ethernet Ethernet:

    DNS-суффикс подключения . . . . . :
    Локальный IPv6-адрес канала . . . : fe80::b6eb:8888:11cc:21ae%10
    IPv4-адрес. . . . . : 192.168.1.13
    Маска подсети . . . . . : 255.255.255.0
    Основной шлюз. . . . . : 192.168.1.1

Адаптер Ethernet Ethernet 2:

    DNS-суффикс подключения . . . . . :
    Локальный IPv6-адрес канала . . . : fe80::7241:9242:3835:ca6f%7
    IPv4-адрес. . . . . : 192.168.56.1
    Маска подсети . . . . . : 255.255.255.0
    Основной шлюз. . . . . :
```

Рис. 1: ipconfig

Адаптер Ethernet Ethernet:

```
DNS-суффикс подключения . . . . . :  
Описание. . . . . : Realtek PCIe GbE Family Controller  
Физический адрес. . . . . : A8-A1-59-9F-24-38  
DHCP включен. . . . . : Да  
Автонастройка включена. . . . . : Да  
Локальный IPv6-адрес канала . . . : fe80::b6eb:8888:11cc:21ae%10(Основной)  
IPv4-адрес. . . . . : 192.168.1.13(Основной)  
Маска подсети . . . . . : 255.255.255.0  
Аренда получена. . . . . : 2 декабря 2025 г. 14:26:02  
Срок аренды истекает. . . . . : 3 декабря 2025 г. 14:26:02  
Основной шлюз. . . . . : 192.168.1.1  
DHCP-сервер. . . . . : 192.168.1.1  
IAID DHCPv6 . . . . . : 111714649  
DUID клиента DHCPv6 . . . . . : 00-01-00-01-2F-E6-16-3A-A8-A1-59-9F-24-38  
DNS-серверы. . . . . : 192.168.1.1  
NetBios через TCP/IP. . . . . : Включен
```

Рис. 2: ipconfig /all

Выбор соединения

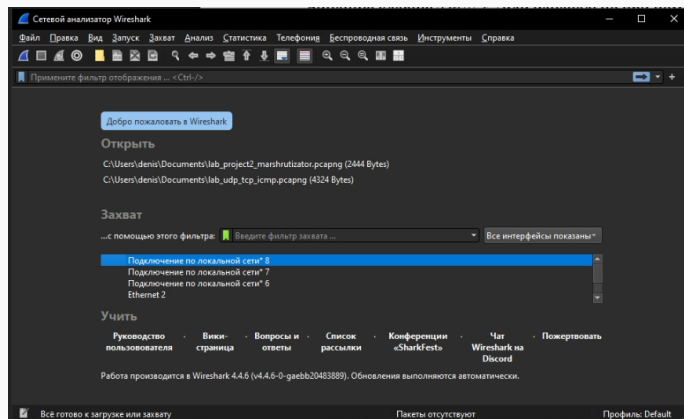


Рис. 3: Выбор соединения

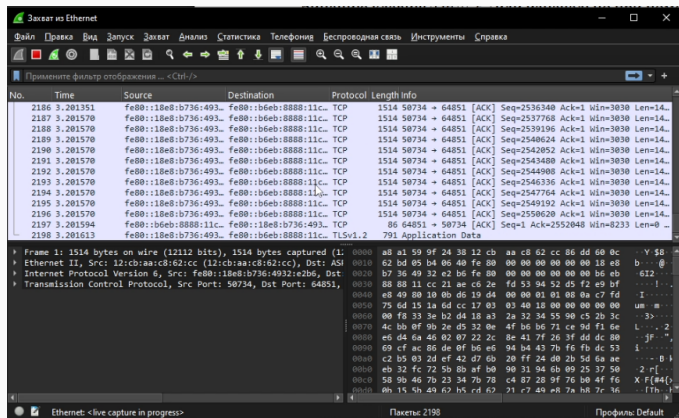


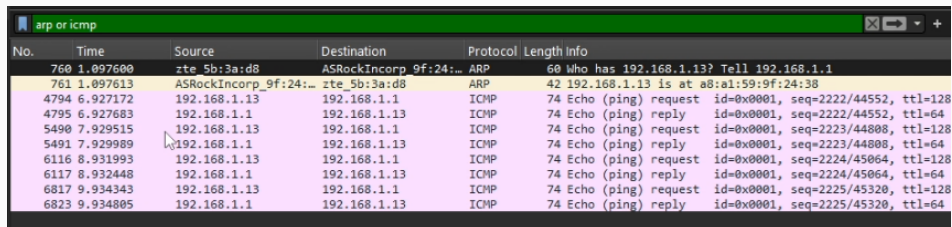
Рис. 4: Wireshark


```
C:\Users\denis>ping 192.168.1.1

Обмен пакетами с 192.168.1.1 по 32 байтами данных:
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64

Статистика Ping для 192.168.1.1:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)
Приблизительное время приема-передачи в мс:
    Минимальное = 0мсек, Максимальное = 0 мсек, Среднее = 0 мсек
```

Рис. 5: ping



No.	Time	Source	Destination	Protocol	Length	Info
760	1.097600	zte 5b:3a:d8	ASRockIncorp 9f:24:...	ARP	60	Who has 192.168.1.13? Tell 192.168.1.1
761	1.097613	ASRockIncorp_9f:24:...	zte_5b:3a:d8	ARP	42	192.168.1.13 is at 08:a1:59:9f:24:38
4794	6.927172	192.168.1.13	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=2222/44552, ttl=128
4795	6.927683	192.168.1.1	192.168.1.13	ICMP	74	Echo (ping) reply id=0x0001, seq=2222/44552, ttl=64
5490	7.929515	192.168.1.13	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=2223/44808, ttl=128
5491	7.929989	192.168.1.1	192.168.1.13	ICMP	74	Echo (ping) reply id=0x0001, seq=2223/44808, ttl=64
6116	8.931993	192.168.1.13	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=2224/45064, ttl=128
6117	8.932448	192.168.1.1	192.168.1.13	ICMP	74	Echo (ping) reply id=0x0001, seq=2224/45064, ttl=64
6817	9.934343	192.168.1.13	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=2225/45320, ttl=128
6823	9.934805	192.168.1.1	192.168.1.13	ICMP	74	Echo (ping) reply id=0x0001, seq=2225/45320, ttl=64

Рис. 6: icmp

Анализ исходящего пакета

```
Epoch Arrival Time: 1764715517.033080000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 0.000093000 seconds]
[Time delta from previous displayed frame: 0.000000000 seconds]
[Time since reference or first frame: 21.201776000 seconds]
Frame Number: 14750
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp || icmpv6]
▼ Ethernet II, Src: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38), Dst: zte_5b:3a:
  ▸ Destination: zte_5b:3a:d8 (34:36:54:5b:3a:d8)
  ▸ Source: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38)
    Type: IPv4 (0x0800)
    [Stream index: 3]
  ▸ Internet Protocol Version 4, Src: 192.168.1.13, Dst: 192.168.1.1
```

Рис. 7: Анализ исходящего пакета

```
Epoch Arrival Time: 1764715517.033561000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 0.000481000 seconds]
[Time delta from previous displayed frame: 0.000481000 seconds]
[Time since reference or first frame: 21.202257000 seconds]
Frame Number: 14751
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp || icmpv6]
▼ Ethernet II, Src: zte_5b:3a:d8 (34:36:54:5b:3a:d8), Dst: ASRockIncorp_9f:24:
  ▸ Destination: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38)
  ▸ Source: zte_5b:3a:d8 (34:36:54:5b:3a:d8)
    Type: IPv4 (0x0800)
    [Stream index: 3]
▼ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.13
```

Рис. 8: Анализ входящего пакета

18_	26.211063	zte_5b:3a:d8	ASRockIncorp_9_	ARP	60 Who has 192.168.1.13?
18_	26.211072	ASRockIncorp_9_	zte_5b:3a:d8	ARP	42 192.168.1.13 is at a8:
63_	91.900394	zte_5b:3a:d8	ASRockIncorp_9_	ARP	60 Who has 192.168.1.13?
63_	91.900406	ASRockIncorp_9_	zte_5b:3a:d8	ARP	42 192.168.1.13 is at a8:
10_	151.949770	zte_5b:3a:d8	ASRockIncorp_9_	ARP	60 Who has 192.168.1.13?
10_	151.949778	ASRockIncorp_9_	zte_5b:3a:d8	ARP	42 192.168.1.13 is at a8:
14_	207.709237	zte_5b:3a:d8	ASRockIncorp_9_	ARP	60 Who has 192.168.1.13?
14_	207.709244	ASRockIncorp_9_	zte_5b:3a:d8	ARP	42 192.168.1.13 is at a8:
19_	258.419003	zte_5b:3a:d8	ASRockIncorp_9_	ARP	60 Who has 192.168.1.13?
19_	258.419014	ASRockIncorp_9_	zte_5b:3a:d8	ARP	42 192.168.1.13 is at a8:
24_	315.638105	zte_5b:3a:d8	ASRockIncorp_9_	ARP	60 Who has 192.168.1.13?
24_	315.638115	ASRockIncorp_9_	zte_5b:3a:d8	ARP	42 192.168.1.13 is at a8:

- Frame 18369: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on in - Section number: 1 - Interface id: 0 (\Device\NPF_{56EA45D7-3010-43F5-943B-4F0074E44FE1}) - Encapsulation type: Ethernet (1) - Arrival Time: Dec 3, 2025 01:45:22.042367000 RTZ 2 (зима) - UTC Arrival Time: Dec 2, 2025 22:45:22.042367000 UTC - Epoch Arrival Time: 1764715522.042367000 [Time shift for this packet: 0.000000000 seconds] [Time delta from previous captured frame: 0.010127000 seconds] [Time delta from previous displayed frame: 2.000812000 seconds] [Time since reference or first frame: 26.211063000 seconds] - Frame Number: 18369 - Frame Length: 60 bytes (480 bits) - Capture Length: 60 bytes (480 bits) [Frame is marked: False] [Frame is ignored: False] [Protocols in frame: eth:ethertype:arp] [Coloring Rule Name: ARP] [Coloring Rule String: arp] - Ethernet II, Src: zte_5b:3a:d8 (34:36:54:5b:3a:d8), Dst: ASRockIncorp_9f:24: - Destination: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38)	00 00 00 00
--	----------------------

Второй arp пакет

18_	26.211063	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	Who has 192.168.1.13? Tell 192.
18_	26.211072	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	192.168.1.13 is at a8:a1:59:9f:
63_	91.900394	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	Who has 192.168.1.13? Tell 192.
63_	91.900406	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	192.168.1.13 is at a8:a1:59:9f:
10_	151.949770	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	Who has 192.168.1.13? Tell 192.
10_	151.949778	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	192.168.1.13 is at a8:a1:59:9f:
14_	207.709237	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	Who has 192.168.1.13? Tell 192.
14_	207.709244	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	192.168.1.13 is at a8:a1:59:9f:
19_	258.419003	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	Who has 192.168.1.13? Tell 192.
19_	258.419014	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	192.168.1.13 is at a8:a1:59:9f:
24_	315.638105	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	Who has 192.168.1.13? Tell 192.
24_	315.638115	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	192.168.1.13 is at a8:a1:59:9f:

Arrival Time: Dec 3, 2025 01:45:22.042376000 RTZ 2 (зима)	0000	34 36
UTC Arrival Time: Dec 2, 2025 22:45:22.042376000 UTC	0010	08 00
Epoch Arrival Time: 1764715522.042376000	0020	34 36
[Time shift for this packet: 0.000000000 seconds]		
[Time delta from previous captured frame: 0.000009000 seconds]		
[Time delta from previous displayed frame: 0.000009000 seconds]		
[Time since reference or first frame: 26.211072000 seconds]		
Frame Number: 18370		
Frame Length: 42 bytes (336 bits)		
Capture Length: 42 bytes (336 bits)		
[Frame is marked: False]		
[Frame is ignored: False]		
[Protocols in frame: eth:ethertype:arp]		
[Coloring Rule Name: ARP]		
[Coloring Rule String: arp]		
↳ Ethernet II, Src: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38), Dst: zte_5b:3a:		
↳ Destination: zte_5b:3a:d8 (34:36:54:5b:3a:d8)		
↳ Source: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38)		
Type: ARP (0x0806)		
[Stream index: 3]		
↳ Address Resolution Protocol (reply)		
↳ Stream index: 4 (eth stream)		

Рис. 10: Второй arp пакет

```
C:\Users\denis>ping yandex.ru

Обмен пакетами с yandex.ru [77.88.44.55] с 32 байтами данных:
Ответ от 77.88.44.55: число байт=32 время=7мс TTL=54
Ответ от 77.88.44.55: число байт=32 время=7мс TTL=54
Ответ от 77.88.44.55: число байт=32 время=7мс TTL=54
Ответ от 77.88.44.55: число байт=32 время=7мс TTL=54

Статистика Ping для 77.88.44.55:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)
Приблизительное время приема-передачи в мс:
    Минимальное = 7мсек, Максимальное = 7 мсек, Среднее = 7 мсек
```

Рис. 11: ping ya.ru

No.	Time	Source	Destination	Protocol	Length	Info
701	0.994405	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	Who has 192.168.1.1? Tell 192.168.1.13
702	0.994838	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	192.168.1.1 is at 34:36:54:5b:3a:d8
91...	13.494326	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	Who has 192.168.1.1? Tell 192.168.1.13
91...	13.494746	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	192.168.1.1 is at 34:36:54:5b:3a:d8
13...	19.994495	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	Who has 192.168.1.1? Tell 192.168.1.13
13...	19.994894	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	192.168.1.1 is at 34:36:54:5b:3a:d8

▼ Frame 701: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{56EA45D7-3010-43F5-943B-4F0074E44FE1}

Section number: 1

► Interface id: 0 (\Device\NPF_{56EA45D7-3010-43F5-943B-4F0074E44FE1})

Encapsulation type: Ethernet (1)

Arrival Time: Dec 3, 2025 02:23:06.888395000 RTZ 2 (зима)

UTC Arrival Time: Dec 2, 2025 23:23:06.888395000 UTC

Epoch Arrival Time: 1764717786.888395000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 0.027341000 seconds]

[Time delta from previous displayed frame: 0.000000000 seconds]

[Time since reference or first frame: 0.994405000 seconds]

Frame Number: 701

Frame Length: 42 bytes (336 bits)

Capture Length: 42 bytes (336 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:arp]

[Coloring Rule Name: ARP]

[Coloring Rule String: arp]

► Ethernet II, Src: ASRockIncorp_9f:24:38 (a8:c1:59:9f:24:38), Dst: zte_5b:3a:d8 (34:36:54:5b:3a:d8)

Рис. 12: arp пакеты

Второй arp пакет

702	0.994838	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	192.168.1.1 is at 34:36:54:5b:3a:d8
91...	13.494326	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	Who has 192.168.1.1? Tell 192.168.1.13
91...	13.494746	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	192.168.1.1 is at 34:36:54:5b:3a:d8
13...	19.994495	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42	Who has 192.168.1.1? Tell 192.168.1.13
13...	19.994894	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60	192.168.1.1 is at 34:36:54:5b:3a:d8

▼ Frame 702: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{56EA45} Section number: 1

- ▶ Interface id: 0 (\Device\NPF_{56EA45D7-3010-43F5-943B-4F0074E44FE1})
- Encapsulation type: Ethernet (1)
- Arrival Time: Dec 3, 2025 02:23:06.888828000 RTZ 2 (зима)
- UTC Arrival Time: Dec 2, 2025 23:23:06.888828000 UTC
- Epoch Arrival Time: 1764717786.888828000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 0.000433000 seconds]
- [Time delta from previous displayed frame: 0.000433000 seconds]
- [Time since reference or first frame: 0.994838000 seconds]
- Frame Number: 702
- Frame Length: 60 bytes (480 bits)
- Capture Length: 60 bytes (480 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:arp]
- [Coloring Rule Name: ARP]
- [Coloring Rule String: arp]
- ▶ Ethernet II, Src: zte_5b:3a:d8 (34:36:54:5b:3a:d8), Dst: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38)
- ▶ Address Resolution Protocol (reply)

Рис. 13: Второй arp пакет

Исходящий пакет icmp

→	57...	773.029123	192.168.1.13	77.88.44.55	ICMP	74 Echo (ping) request
←	57...	773.036635	77.88.44.55	192.168.1.13	ICMP	74 Echo (ping) reply
	57...	774.031942	192.168.1.13	77.88.44.55	ICMP	74 Echo (ping) request
	57...	774.039372	77.88.44.55	192.168.1.13	ICMP	74 Echo (ping) reply
	57...	776.023446	zte_5b:3a:d8	ASRockIncorp_9...	ARP	60 Who has 192.168.1.13
	57...	776.023463	ASRockIncorp_9...	zte_5b:3a:d8	ARP	42 192.168.1.13 is at a

Encapsulation type: Ethernet (1)
Arrival Time: Dec 3, 2025 01:57:48.860427000 RTZ 2 (зима)
UTC Arrival Time: Dec 2, 2025 22:57:48.860427000 UTC
Epoch Arrival Time: 1764716268.860427000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 0.029605000 seconds]
[Time delta from previous displayed frame: 0.994584000 seconds]
[Time since reference or first frame: 773.029123000 seconds]
Frame Number: 570663
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp icmpv6]
▼ Ethernet II, Src: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38), Dst: zte_5b:3a:
► Destination: zte_5b:3a:d8 (34:36:54:5b:3a:d8)
► Source: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38)
Type: IPv4 (0x0800)
[Stream index: 3]

Рис. 14: Исходящий пакет icmp

Входящий пакет icmp

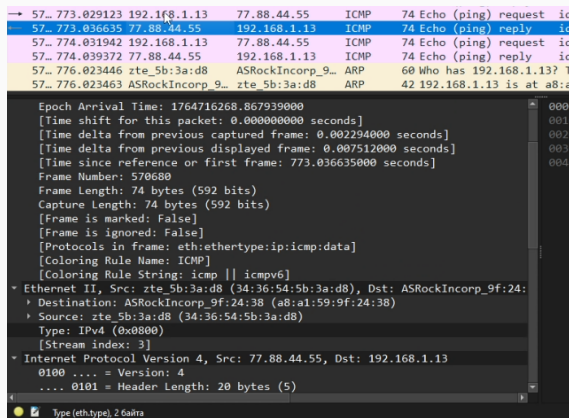


Рис. 15: Входящий пакет icmp



Рис. 16: Подключение по http

Исходящий http пакет

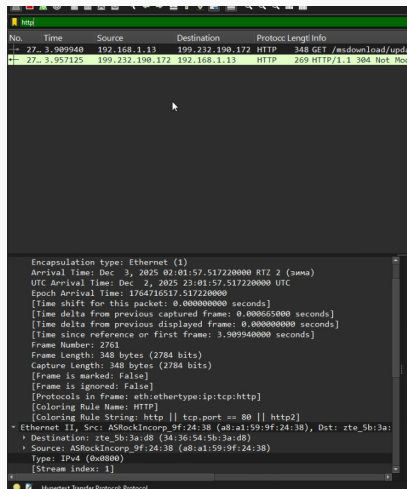


Рис. 17: Исходящий http пакет

Входящий http пакет

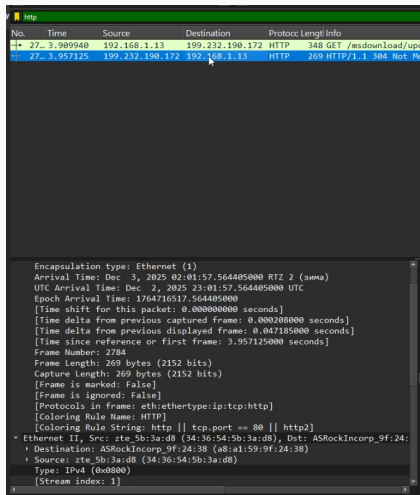


Рис. 18: Входящий http пакет

Исходящий dns пакет

```
87.. 125.077182 192.168.1.13 192.168.1.1 DNS 81 Standard query 0x9c5d HT
87.. 125.077337 192.168.1.13 192.168.1.1 DNS 81 Standard query 0xbd5 A
87.. 125.082873 192.168.1.13 192.168.1.1 DNS 71 Standard query 0xfe04 HT
87.. 125.082981 192.168.1.13 192.168.1.1 DNS 71 Standard query 0xa0e6 A
87.. 125.083152 192.168.1.13 192.168.1.1 DNS 84 Standard query 0x7056 HT
87.. 125.083237 192.168.1.13 192.168.1.1 DNS 84 Standard query 0x3cb8 A
87.. 125.083629 108.177.14.95 192.168.1.13 QUIC 66 Protected Payload (KP0)
87.. 125.099053 64.233.161.93 192.168.1.13 QUIC 66 Protected Payload (KP0)
87.. 125.099453 192.168.1.13 192.168.1.1 DNS 81 Standard query 0xb436 HT
87.. 125.099603 192.168.1.13 192.168.1.1 DNS 81 Standard query 0x9eae A
87.. 125.104400 192.168.1.13 192.168.1.1 DNS 71 Standard query 0xc7fe HT
87.. 125.104534 192.168.1.13 192.168.1.1 DNS 71 Standard query 0x873a A
87.. 125.104637 192.168.1.13 192.168.1.1 DNS 84 Standard query 0xe8ad HT
87.. 125.104723 192.168.1.13 192.168.1.1 DNS 84 Standard query 0xa06f A
87.. 125.107235 64.233.162.106 192.168.1.13 QUIC 66 Protected Payload (KP0)

> 000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: UDP (17)
Header Checksum: 0x17e2 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.13
Destination Address: 192.168.1.1
[Stream index: 4]
> User Datagram Protocol, Src Port: 54573, Dst Port: 53
Source Port: 54573
Destination Port: 53
Length: 47
Checksum: 0x504e [unverified]
[Checksum Status: Unverified]
[Stream index: 68]
[Stream Packet Number: 1]
> [Timestamps]
UDP payload (39 bytes)
> Domain Name System (query)
```

Рис. 19: Исходящий dns пакет

Входящий dns пакет

86...	124.751930	192.168.1.1	192.168.1.13	QUIC	70 Protected Payload (KP0)
86...	124.751654	192.168.1.1	192.168.1.13	DNS	169 Standard query response 0x63ca HTTPS home
86...	124.751930	192.168.1.1	192.168.1.13	DNS	158 Standard query response 0x809a A home.web
86...	124.752127	192.168.1.1	192.168.1.13	DNS	169 Standard query response 0xa8f9 HTTPS home
86...	124.752321	192.168.1.1	192.168.1.13	DNS	158 Standard query response 0xc7ef A home.web
86...	124.752518	192.168.1.1	192.168.1.13	DNS	169 Standard query response 0x358b HTTPS home
86...	124.752723	192.168.1.1	192.168.1.13	DNS	158 Standard query response 0x61a7 A home.web
86...	124.752929	192.168.1.1	192.168.1.13	DNS	158 Standard query response 0xee7a A home.web
86...	124.753136	192.168.1.1	192.168.1.13	DNS	169 Standard query response 0x6a5c HTTPS home
86...	124.753652	192.168.1.1	192.168.1.13	DNS	158 Standard query response 0xc913 A home.web
86...	124.769075	192.168.1.13	108.177.14.95	QUIC	74 Protected Payload (KP0), DCID=e6094f9ca6f
86...	124.775132	192.168.1.13	64.233.161.93	QUIC	74 Protected Payload (KP0), DCID=f52f6a4fb61
86...	124.785133	108.177.14.95	192.168.1.13	QUIC	679 Protected Payload (KP0)
86...	124.785504	108.177.14.95	192.168.1.13	QUIC	162 Protected Payload (KP0)
86...	124.785754	192.168.1.13	108.177.14.95	QUIC	77 Protected Payload (KP0), DCID=e6094f9ca6f
86...	124.791835	108.177.14.95	192.168.1.13	QUIC	645 Protected Payload (KP0)

Frame 86788: 169 bytes on wire (1352 bits), 169 bytes captured (1352 bits) on interface \Device\NPF...
Section number: 1
Interface id: 0 (\Device\NPF_{56EA45D7-3010-43F5-943B-4F0074E44FE1})
Encapsulation type: Ethernet (1)
Arrival Time: Dec 3, 2025 02:03:58.358934000 RTZ 2 (зима)
UTC Arrival Time: Dec 2, 2025 23:03:58.358934000 UTC
Epoch Arrival Time: 1764716638.358934000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 0.002413000 seconds]
[Time delta from previous displayed frame: 0.002413000 seconds]
[Time since reference or first frame: 124.751654000 seconds]
Frame Number: 86788
Frame Length: 169 bytes (1352 bits)
Capture Length: 169 bytes (1352 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:dns]
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: zte_5b:3a:d8 (34:36:54:5b:3a:d8), Dst: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38)
Destination: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38)

Рис. 20: Входящий dns пакет

Исходящий quic пакет

87...	125.083629	108.177.14.95	192.168.1.13	QUIC	66 Protected Payload (KP0)
87...	125.099053	64.233.161.93	192.168.1.13	QUIC	66 Protected Payload (KP0)
87...	125.099453	192.168.1.13	192.168.1.1	DNS	81 Standard query 0xb436 HTTPS framework.web
87...	125.099603	192.168.1.13	192.168.1.1	DNS	81 Standard query 0x9eae A framework.web.cer
87...	125.104400	192.168.1.13	192.168.1.1	DNS	71 Standard query 0xc7fe HTTPS cds.cern.ch
87...	125.104534	192.168.1.13	192.168.1.1	DNS	71 Standard query 0x873a A cds.cern.ch
87...	125.104637	192.168.1.13	192.168.1.1	DNS	84 Standard query 0xe8ad HTTPS webanalytics.
87...	125.104723	192.168.1.13	192.168.1.1	DNS	84 Standard query 0xa06f A webanalytics.web.
87...	125.107235	64.233.162.106	192.168.1.13	QUIC	66 Protected Payload (KP0)

▼ Frame 87074: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{56EA45D7-3010-43F5-943B-4F0074E44FE1}

Section number: 1

► Interface id: 0 (\Device\NPF_{56EA45D7-3010-43F5-943B-4F0074E44FE1})

Encapsulation type: Ethernet (1)

Arrival Time: Dec 3, 2025 02:03:58.690909000 RTZ 2 (зима)

UTC Arrival Time: Dec 2, 2025 23:03:58.690909000 UTC

Epoch Arrival Time: 1764716638.690909000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 0.000392000 seconds]

[Time delta from previous displayed frame: 0.000392000 seconds]

[Time since reference or first frame: 125.083629000 seconds]

Frame Number: 87074

Frame Length: 66 bytes (528 bits)

Capture Length: 66 bytes (528 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:udp:quic]

[Coloring Rule Name: UDP]

[Coloring Rule String: udp]

▼ Ethernet II, Src: zte_5b:3a:d8 (34:36:54:5b:3a:d8), Dst: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38)

► Destination: ASRockIncorp_9f:24:38 (a8:a1:59:9f:24:38)

QUIC IETF (quic), 24 bytes

Пакеты: 533

Рис. 21: Исходящий quic пакет

Входящий quic пакет

```
90... 126.188725 192.168.1.13 74.125.131.95 QUIC 1292 Initial, DCID=831a1f0852f2ae4f, PKN:
90... 126.188766 192.168.1.13 74.125.131.95 QUIC 1292 Initial, DCID=831a1f0852f2ae4f, PKN:
90... 126.188793 192.168.1.13 74.125.131.95 QUIC 122 0-RTT, DCID=831a1f0852f2ae4f
90... 126.205623 74.125.131.95 192.168.1.13 QUIC 82 Initial, SCID=e31a1f0852f2ae4f, PKN:
90... 126.205834 74.125.131.95 192.168.1.13 QUIC 1292 Initial, SCID=e31a1f0852f2ae4f, PKN:
90... 126.206508 74.125.131.95 192.168.1.13 QUIC 1292 Protected Payload (KP0)
90... 126.206508 74.125.131.95 192.168.1.13 QUIC 848 Protected Payload (KP0)
90... 126.206508 74.125.131.95 192.168.1.13 QUIC 242 Protected Payload (KP0)
90... 126.206813 192.168.1.13 74.125.131.95 QUIC 120 Handshake, DCID=e31a1f0852f2ae4f
90... 126.206847 192.168.1.13 74.125.131.95 QUIC 73 Protected Payload (KP0), DCID=e31a1f08
90... 126.206989 192.168.1.13 74.125.131.95 QUIC 1288 Protected Payload (KP0), DCID=e31a1f08
90... 126.207016 192.168.1.13 74.125.131.95 QUIC 269 Protected Payload (KP0), DCID=e31a1f08

[Timestamps]
UDP payload (1250 bytes)
QUIC IETF
  QUIC Connection information
    [Packet Length: 1250]
    1... .... = Header Form: Long Header (1)
    .1.. .... = Fixed Bit: True
    ..00 .... = Packet Type: Initial (0)
    [.... 00.. = Reserved: 0]
    [.... ..00 = Packet Number Length: 1 bytes (0)]
    Version: 1 (0x00000001)
    Destination Connection ID Length: 8
    Destination Connection ID: 831a1f0852f2ae4f
    Source Connection ID Length: 0
    Token Length: 70
    Token: 008c543a4a92b05aa249fa81a077408f36c0cf54b09965da55ecc54bb4407c5ee99647e6a2d803c226d9d54
    Length: 1161
    [Packet Number: 1]
    Payload [...]: fc296a41202c49e0c51f197088060a119a64f3ed8e5ffb0599b6e66a62bb968069d3ab524f67e535eb
  CRYPTO
```

Рис. 22: Входящий quic пакет

```

39_ 310.479206 fe80::18e8:b73_ fe80::b6eb:888... TLSv... 791 Application Data
39_ 310.479206 fe80::18e8:b73_ fe80::b6eb:888... TCP 1514 50734 → 64851 [ACK] Seq=247366140 Ack=10
39_ 310.479206 fe80::18e8:b73_ fe80::b6eb:888... TCP 1514 50734 → 64851 [ACK] Seq=247367568 Ack=10
39_ 310.479206 fe80::18e8:b73_ fe80::b6eb:888... TCP 1514 50734 → 64851 [ACK] Seq=247368996 Ack=10
39_ 310.479206 fe80::18e8:b73_ fe80::b6eb:888... TCP 1514 50734 → 64851 [ACK] Seq=247370424 Ack=10
39_ 310.479206 fe80::18e8:b73_ fe80::b6eb:888... TLSv... 229 Application Data
39_ 310.479221 fe80::b6eb:888_ fe80::18e8:b73... TCP 86 64851 → 50734 [ACK] Seq=100045 Ack=24737
39_ 310.512343 fe80::18e8:b73_ fe80::b6eb:888... TCP 1514 50734 → 64851 [ACK] Seq=247371995 Ack=10
39_ 310.512890 fe80::18e8:b73_ fe80::b6eb:888... TCP 1514 50734 → 64851 [ACK] Seq=247373423 Ack=10
39_ 310.512890 fe80::18e8:b73_ fe80::b6eb:888... TCP 1514 50734 → 64851 [ACK] Seq=247374851 Ack=10
39_ 310.512890 fe80::18e8:b73_ fe80::b6eb:888... TCP 1514 50734 → 64851 [ACK] Seq=247376279 Ack=10
39_ 310.512890 fe80::18e8:b73_ fe80::b6eb:888... TCP 1514 50734 → 64851 [ACK] Seq=247377707 Ack=10

> Internet Protocol Version 6, Src: fe80::18e8:b736:4932:e2b6, Dst: fe80::b6eb:8888:11cc:21ae
> Transmission Control Protocol, Src Port: 50734, Dst Port: 64851, Seq: 247366140, Ack: 100045, Len: :
    Source Port: 50734
    Destination Port: 64851
    [Stream index: 0]
    [Stream Packet Number: 213973]
    [Conversation completeness: Incomplete (12)]
    [TCP Segment Len: 1428]
    Sequence Number: 247366140 (relative sequence number)
    Sequence Number (raw): 219584296
    [Next Sequence Number: 247367568 (relative sequence number)]
    Acknowledgment Number: 100045 (relative ack number)
    Acknowledgment number (raw): 3922144156
    1000 .... = Header Length: 32 bytes (8)
    > Flags: 0x010 (ACK)
    Window: 3030
    [Calculated window size: 3030]
    [Window size scaling factor: -1 (unknown)]
    Checksum: 0xea35 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0

```

```
Sequence Number: 0      (relative sequence number)
Sequence Number (raw): 1484550668
[Next Sequence Number: 1      (relative sequence number)]
Acknowledgment Number: 1      (relative ack number)
Acknowledgment number (raw): 1476315037
1000 .... = Header Length: 32 bytes (8)
▼ Flags: 0x012 (SYN, ACK)
  000. .... = Reserved: Not set
  ...0 .... = Accurate ECN: Not set
  .... 0... = Congestion Window Reduced: Not set
  .... .0.. = ECN-Echo: Not set
  .... ..0. = Urgent: Not set
  .... ...1 .... = Acknowledgment: Set
  .... .... 0... = Push: Not set
  .... .... .0.. = Reset: Not set
  ► .... .... ..1. = Syn: Set
  .... .... ...0 = Fin: Not set
  [TCP Flags: .....A..S.]
Window: 32120
[Calculated window size: 32120]
Checksum: 0x43d3 [unverified]
[Checksum Status: Unverified]
```

```

Sequence Number (raw): 1478519057
[Next Sequence Number: 1    (relative sequence number)]
Acknowledgment Number: 1    (relative ack number)
Acknowledgment number (raw): 1484550669
0101 .... = Header Length: 20 bytes (5)
- Flags: 0x010 (ACK)
  000. .... = Reserved: Not set
  ...0 .... = Accurate ECN: Not set
  .... 0... = Congestion Window Reduced: Not set
  .... .0.. = ECN-Echo: Not set
  .... ..0. = Urgent: Not set
  .... ...1 = Acknowledgment: Set
  .... .... 0... = Push: Not set
  .... .... .0.. = Reset: Not set
  .... .... ..0. = Syn: Not set
  .... .... ...0 = Fin: Not set
  [TCP Flags: .....A....]
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]

```

Хендшейк на графике потока

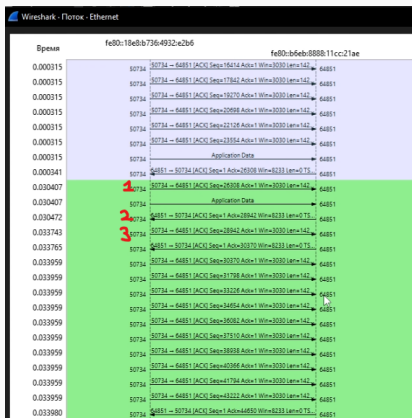


Рис. 26: Хендшейк на графике потока

В результате выполнения работы были получены навыки анализа пакетов в wireshark